



Article Cloud-Base Architecture for Wholesale World.

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1. Introduction

Wholesale World is a prominent global retailer that operates on a membership-based model, delivering an extensive range of high-quality products at remarkably competitive prices. The company has earned a reputation for its efficient and well-structured supply chain, which supports its bulk sales approach and ensures consistent value for its members. With a strong commitment to customer satisfaction, Wholesale World goes beyond just offering products, providing exceptional member-focused services that enhance the overall shopping experience. Driven by a mission to create lasting value, the company emphasizes innovation in its operations and remains dedicated to sustainable growth, positioning itself as a leader in the retail industry.

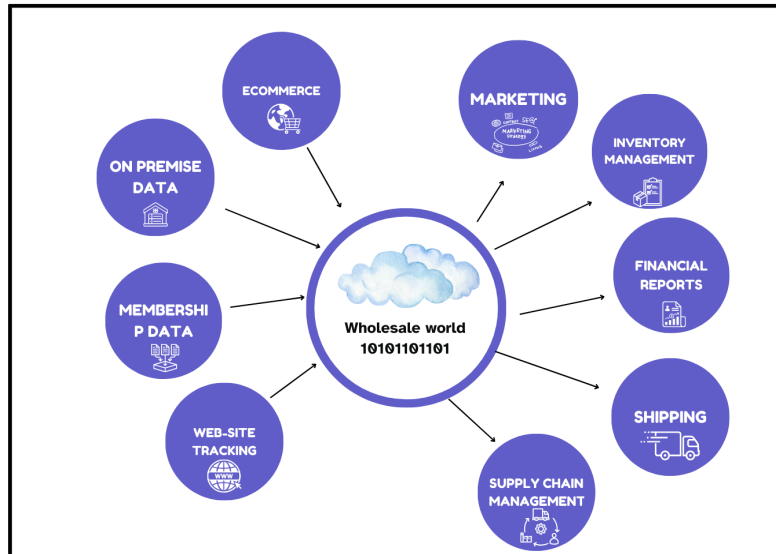
2. Mission

Our mission is to streamline and optimize data flow processes within wholesale warehouses to deliver real-time, accurate, and actionable insights. By leveraging advanced technologies and efficient practices, we aim to enhance operational efficiency, improve decision-making, and elevate customer satisfaction. Through this commitment, we strive to drive sustainable growth and set a benchmark for innovation and excellence in the wholesale industry.

3. Objectives

- I. **Optimizing Inventory Management:** Harness real-time data analytics to prevent overstocking or stockouts, ensuring smooth and efficient supply chain operations across all levels.
- II. **Enhancing Member Experience:** Utilize predictive analytics to deliver personalized recommendations and elevate customer service quality, fostering stronger member satisfaction and loyalty.
- III. **Supporting Data-Driven Decisions:** Equip leadership with comprehensive insights into sales trends, operational performance, and financial health, enabling informed and strategic decision-making.
- IV. **Scalable Infrastructure Development:** Implement robust cloud-based solutions designed to efficiently manage growing data volumes and increasing complexity, ensuring long-term adaptability and reliability.
- V. **Driving Cost Efficiency:** Reduce cloud-related expenses by leveraging reusable datasets and scalable infrastructure, optimizing resource utilization while maintaining operational excellence.

4. Vision Diagram



5. Data Source.

I. On-Premises Data:

This type of data includes transactional, numeric, and categorical information. It is typically processed in batches, either daily or periodically, for aggregation and reporting. On-premises data is essential for managing inventory, tracking sales performance, and driving personalized marketing campaigns and promotions.

II. Membership Details:

Membership data consists of category, numeric, and date/time

information. It is updated periodically through batch processing, particularly for billing or reporting purposes. This data stores customer profiles, membership levels, renewal history, and contact details, helping to manage member engagement effectively.

III. Website Tracking:

Website tracking data includes numeric, categorical, and event-based information, processed through batch processing. It captures user behavior on the Wholesale World website, including page visits, navigation paths, and session durations. This semi-structured data is crucial for analyzing user engagement and understanding customer interactions with the site.

IV. E-Commerce Data:

E-commerce data consists of transactional, numeric, and categorical information, including customer details, product data, and order information. It is processed as streaming data, allowing for real-time updates. This data supports inventory and supply chain management, tracks sales performance, forecasts revenue, and helps analyze customer behavior to enhance the overall user experience.

6. Data Sink.

I. Financial:

This data sink integrates feedback and transaction data to analyze revenue, costs, and profit margins. It provides valuable insights for budgeting, financial reporting, and profitability analysis. Financial reports are delivered using Power BI, offering an efficient way to track and visualize financial performance.

II. Inventory Management:

By leveraging on-premises data and membership details, inventory management supports accurate stock forecasting and shelf

replenishment. It helps minimize overstocking or stockouts, ensuring optimal inventory levels. Power BI is used to generate detailed inventory level reports for better decision-making and stock control.

III. Supply Chain Operations:

This sink uses sales and inventory data to optimize replenishment and distribution, enabling efficient tracking of stock levels and demand planning. Reports are generated to provide real-time insights into supply chain performance, ensuring smooth operations and timely restocking.

IV. Marketing:

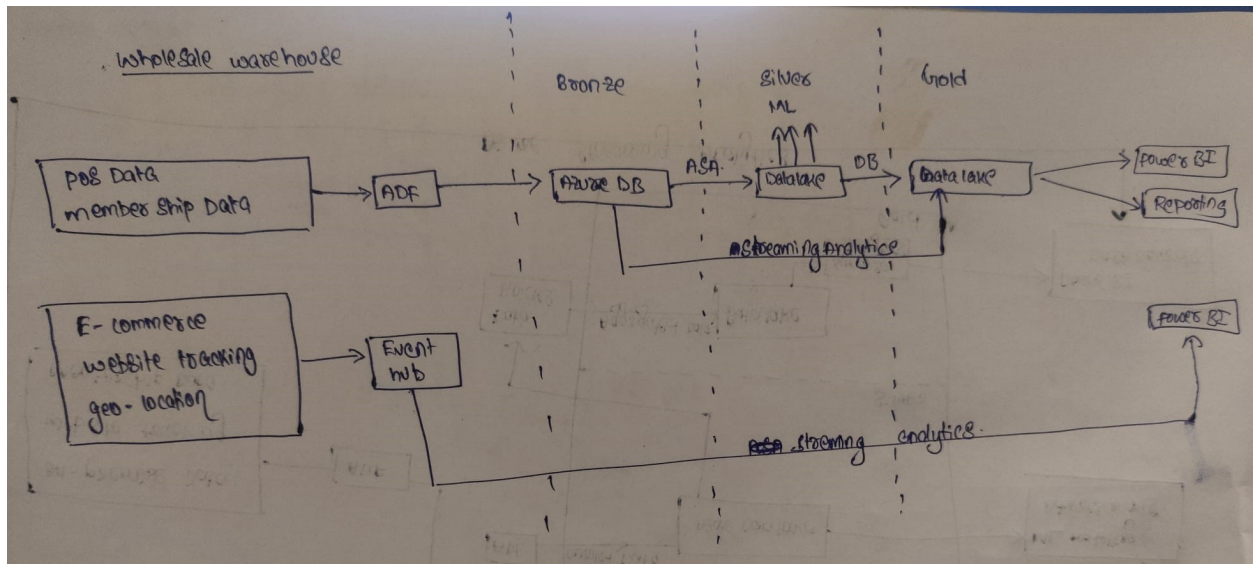
Marketing efforts are driven by sales, e-commerce, and membership data, which are used to create targeted promotions. These promotions are delivered through various digital channels such as email, social media, push notifications, and SMS, ensuring effective customer engagement and campaign reach.

V. Shipping:

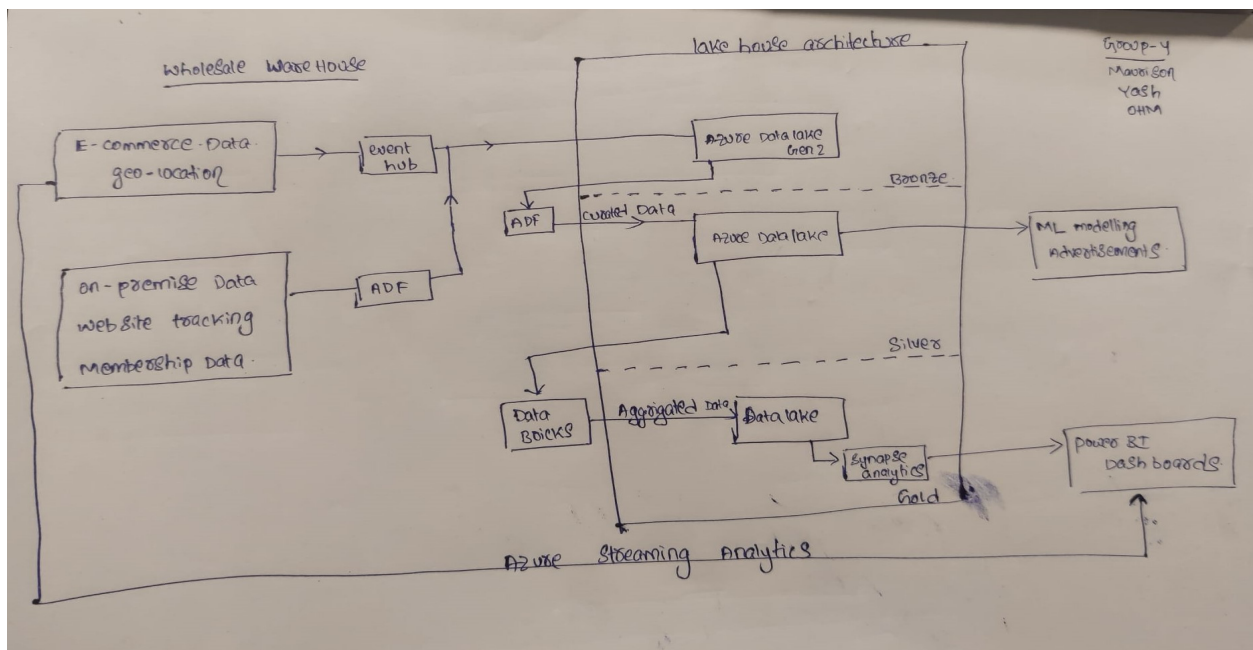
E-commerce data is utilized to manage shipping information, which is outsourced to third-party contractors. Live dashboards provide real-time tracking and updates, ensuring efficient shipment monitoring and timely deliveries.

7. Cloud Architecture.

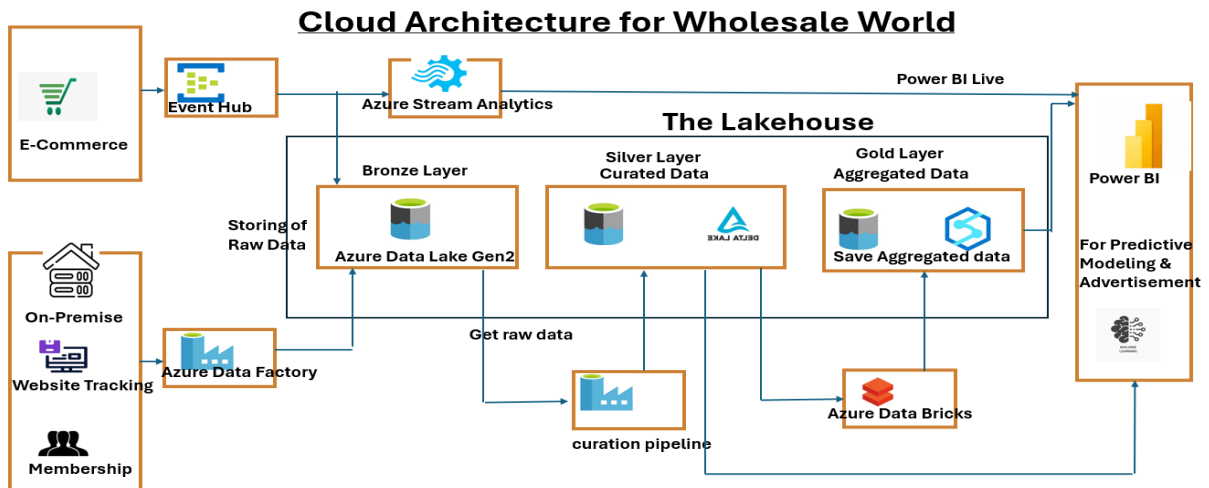
(Phase 1)



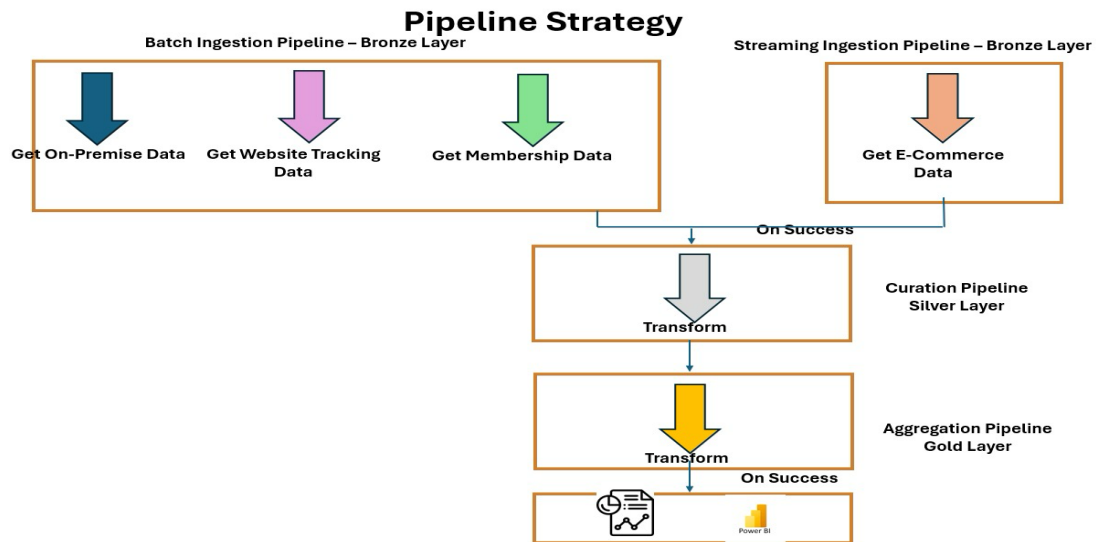
(Phase 2)



Final Cloud Architecture:



8. Pipeline strategy:



The pipeline strategy outlines a comprehensive end-to-end data flow process, emphasizing efficient data ingestion, transformation, storage, and consumption. Here's a detailed breakdown of the approach:

I. Data Ingestion:

Data is captured in both real-time and batch formats from various sources, including e-commerce transactions, website tracking, and sales databases. Tools like Event Hub and Azure Data Factory are used to collect and transfer the data to the storage layer, ensuring both scalability and reliability.

II. Raw Data Storage (Bronze Layer):

Ingested data is initially stored in its raw format within the Bronze Layer. This layer serves as a repository for audit and archival purposes, ensuring that the original data remains intact for traceability.

III. Data Transformation (Curation Pipeline):

The raw data undergoes a cleaning and transformation process using Azure Databricks. This process generates curated datasets, which are stored in the Silver Layer, making them ready for further processing or analysis. Delta Lake plays a crucial role in managing schemas and enabling ACID transactions, ensuring data consistency and reliability.

IV. Data Aggregation and Analysis (Gold Layer):

In the Gold Layer, curated data is further aggregated and analyzed using Azure Synapse Analytics to generate business-ready insights. This step optimizes the data for easy consumption by downstream systems and ensures that it is aligned with business objectives.

V. Data Consumption:

Processed data is delivered to Power BI dashboards from the Gold Layer, and machine learning models from the Silver Layer, to facilitate visualization, predictive analytics, and informed decision-making. The pipeline supports both scheduled and real-time updates, ensuring timely and actionable insights for stakeholders.

9. Conclusion:

The cloud architecture for Wholesale World offers a scalable and efficient data-driven solution designed to optimize inventory management, improve customer experience, and facilitate real-time decision-making. By leveraging Azure services, the system ensures smooth integration, powerful analytics, and cost-effective operations, allowing Wholesale World to achieve operational excellence, foster sustainable growth, and maintain flexibility in its wholesale and retail operations.